

Positive-Real Asymptotic Inversion

with Powers and Logarithms

A constructive coefficient theory
and a Wolfram Language package

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Abstract

We construct the positive real inverse near zero of

$$f(x) = ax^p \left(1 + \sum_{j=1}^m x^{\delta_j} P_j(\log x) \right), \quad a, p, \delta_j > 0,$$

where the P_j are real polynomials and the gaps δ_j may be irrational. A homotopy form of Lagrange–Bürmann inversion produces every coefficient by a finite product of first-order differential operators on polynomials. We prove local existence, branch selection, convergence of the polynomial-block expansion, formal uniqueness, and logarithm-aware remainder estimates for both exponent and perturbation-depth truncation. The original examples $x + x^2(1 + \log x)$ and $x + x^{\sqrt{2}}$ are worked out to arbitrary order, with explicit coefficients and convergence information. The accompanying package normalizes admissible input, preserves exact irrational exponents, combines coincident weights, and returns expressions together with error metadata.

Verification status. The mathematical specification passed 155 independent executed checks, including formal substitution and 100-digit numerical comparisons. The Wolfram source and 35 native regression tests are supplied, but native kernel execution was unavailable in this session. No native test success is claimed.

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1 The question, the answer, and the correct real branch

The archived question asks for an asymptotic inverse on the positive real axis, not for an arbitrary local complex inverse. Its principal example is

$$f(x) = x + x^2(1 + \log x), \quad x > 0, \quad (1)$$

and its second example is $x + x^{\sqrt{2}}$. The reported computer-algebra attempts sometimes remain unevaluated and sometimes select a non-real inverse branch [1]. The mathematical issue is not resolved merely by appending $y > 0$ to an otherwise unspecified inverse: one must choose the inverse germ whose argument tends to zero along the positive real axis.

For (1), the answer begins

$$\begin{aligned} g(y) = & y - y^2(1 + \log y) + y^3(2 \log^2 y + 5 \log y + 3) \\ & - y^4 \left(5 \log^3 y + \frac{41}{2} \log^2 y + 27 \log y + \frac{23}{2} \right) + O(y^5 |\log y|^4). \end{aligned} \quad (2)$$

All limits in this report are $y \rightarrow 0^+$ unless another regime is explicitly specified. The exact all-orders coefficient formula is given in [section 7](#); it is not inferred by numerical fitting.

For the irrational-power example the answer is

$$\begin{aligned} (x + x^{\sqrt{2}})^{-1}(y) = & y - y^{\sqrt{2}} + \sqrt{2} y^{2\sqrt{2}-1} - \frac{6 - \sqrt{2}}{2} y^{3\sqrt{2}-2} \\ & + \left(\frac{17\sqrt{2}}{3} - 4 \right) y^{4\sqrt{2}-3} + O(y^{5\sqrt{2}-4}). \end{aligned} \quad (3)$$

Here f^{-1} means functional inversion, not a reciprocal. The first four terms reproduce the expected result in the supplied archive, and the next term is generated by the same rule as every subsequent term.

1.1 Why the positive inverse exists globally in the two examples

For (1),

$$f'(x) = 1 + x(3 + 2 \log x).$$

The derivative of $x(3 + 2 \log x)$ is $5 + 2 \log x$, so its global minimum occurs at $x = e^{-5/2}$. Consequently

$$f'(x) \geq m_0 := 1 - 2e^{-5/2} > 0 \quad (x > 0). \quad (4)$$

Also $f(x) \rightarrow 0$ as $x \rightarrow 0^+$ and $f(x) \rightarrow +\infty$ as $x \rightarrow +\infty$. Thus f is a bijection of $(0, \infty)$ onto itself. Its inverse satisfies $g(y) \sim y$. The sign of the first correction is useful as a check: for small y , $1 + \log y < 0$, and therefore $g(y) > y$.

For $f_\alpha(x) = x + x^\alpha$, with any fixed $\alpha > 1$,

$$f'_\alpha(x) = 1 + \alpha x^{\alpha-1} \geq 1,$$

so the same global positive bijection conclusion holds. In this case $g_\alpha(y) < y$. Both inequalities agree with the expansions.

The general model studied below need only be monotone sufficiently near zero. Higher-power terms with negative coefficients can destroy global monotonicity without affecting the positive local inverse selected by the package.

1.2 Why an ordinary Taylor branch can be the wrong object

Away from the logarithmic cut, the nonzero complex solutions of $f(z) = 0$ in (1) obey

$$z(1 + \log z) = -1.$$

Writing $w = 1 + \log z$ gives $we^w = -e$. Thus such roots can be described, with compatible logarithm branches, by

$$z_0 = \exp(W_k(-e) - 1).$$

At one of these roots $f'(z_0) = z_0 - 1 \neq 0$, so a perfectly legitimate analytic inverse about $y = 0$ has the expansion

$$z_0 + \frac{y}{z_0 - 1} - \frac{3 + 2W_k(-e)}{2(z_0 - 1)^3}y^2 + \dots$$

It is not the positive inverse tending to zero. This calculation explains the shape of the non-real expansion reported in the archive; it does not diagnose a current-version software bug.

The desired inverse is not analytic at the endpoint. Indeed, $f''(x) = 5 + 2 \log x \rightarrow -\infty$, and

$$g''(y) = -\frac{f''(g(y))}{f'(g(y))^3} \rightarrow +\infty.$$

It cannot have an ordinary Taylor expansion at zero. The failure of ordinary Taylor analyticity does not preclude a precise, convergent expansion in power–logarithm blocks.

1.3 What is contributed here

The coefficient identity is a specialization of classical Lagrange–Bürmann reversion, not a claim of a new general inversion theorem. Classical inverse-function and residue formulations are recorded in [2]. The constructive contribution is to normalize a broad class of positive real germs, express its coefficients entirely through finite polynomial operations, justify exact irrational-weight truncation, and implement the resulting specification with explicit failure modes.

The supplied repository is useful methodological context. Its directory README and inspected portions of the companion monograph emphasize choosing a dominant inverse core, organizing perturbations by weight, and transporting a forward residual through a slope estimate [3, 4]. Here these ideas are specialized to a class for which all analytic assertions can be proved directly. No uninspected theorem in a larger transseries monograph is used as a premise.

2 Power–logarithm scales and admissible models

Definition 2.1 (Admissible finite model). An admissible model is

$$f(x) = ax^p(1 + R(x)), \quad R(x) = \sum_{j=1}^m x^{\delta_j} P_j(\log x), \quad (5)$$

where $a > 0$, $p > 0$, each $\delta_j > 0$, and each P_j is a nonzero real polynomial. The number m is finite; $m = 0$ is allowed and gives a pure monomial. All parameters are fixed. Powers and logarithms are real on $x > 0$, and their complex continuations are taken on small disks avoiding the nonpositive real axis when needed in proofs.

The input class includes polynomial perturbations, rational Puiseux powers, exact irrational powers, arbitrary finite polynomial degrees in $\log x$, and mixtures of these features. Coefficients of the P_j may

have either sign. Repeated gaps are permitted mathematically and may be combined by adding their polynomials.

Put

$$t = \left(\frac{y}{a}\right)^{1/p}, \quad L = \log t, \quad T = 1 + |L|. \quad (6)$$

Then $t > 0$ and $t \rightarrow 0^+$ exactly when $y \rightarrow 0^+$. The inverse sought is $g(y) = t(1 + o(1))$.

2.1 The hierarchy of powers and logarithms

For every fixed $\varepsilon > 0$ and real M ,

$$t^\varepsilon (1 + |\log t|)^M \rightarrow 0. \quad (7)$$

One proof is to write $t = e^{-u}$: an exponential decay beats any fixed power of u . It follows that any finite logarithmic polynomial at a larger power of t is smaller than any nonzero logarithmic polynomial at a smaller power. Within a fixed power, the term of highest logarithmic degree dominates in absolute value.

This is why exponent blocks are natural:

$$t^\beta Q(L), \quad Q \in \mathbb{R}[L].$$

A block is kept as a whole. There is no need to rank its finitely many logarithmic monomials separately in the implementation.

A crucial error distinction is

$$t^\beta \log^2 t \notin O(t^\beta), \quad t^\beta \log^2 t \in O(t^{\beta-\varepsilon}) \quad \text{for every } \varepsilon > 0.$$

An order term that suppresses the logarithms can therefore assert a false error bound even when the displayed coefficients are right.

2.2 The positive additive support

For $k = (k_1, \dots, k_m) \in \mathbb{N}_0^m$, write

$$|k| = \sum_j k_j, \quad k! = \prod_j k_j!, \quad \Delta_k = \sum_j k_j \delta_j.$$

The generated weight set is

$$S = \{\Delta_k : k \in \mathbb{N}_0^m\}.$$

If $d = \min_j \delta_j > 0$, then $\Delta_k \geq d|k|$. Thus there are only finitely many multiindices with weight below any prescribed finite cutoff. In particular, S is locally finite, even when its generators are irrational.

It is important not to confuse this positive semigroup with the additive group generated by the same numbers using negative integer coefficients. That group can be dense, while the positive semigroup relevant here has only finitely many points below any bound.

Distinct multiindices can yield the same weight. For example, gaps 1 and 2 give $\Delta_{(2,0)} = \Delta_{(0,1)} = 2$. The coefficient at weight 2 is the sum of both contributions. Declaring the two terms to be different asymptotic scales would be incorrect. Conversely, replacing irrational gaps by decimal approximations can create spurious equalities. The package does neither.

2.3 Existence of the general positive inverse

Proposition 2.2. *For an admissible model, $R(x) = o(1)$ and $xR'(x) = o(1)$. Consequently*

$$f'(x) = apx^{p-1}(1 + o(1)),$$

and f has a unique positive inverse for all sufficiently small $y > 0$. Moreover $g(y) \sim t$ with t defined by (6).

Proof. Every $x^{\delta_j} P_j(\log x)$ tends to zero by (7). Also

$$x \frac{d}{dx} (x^{\delta_j} P_j(\log x)) = x^{\delta_j} (\delta_j P_j(\log x) + P'_j(\log x)) = o(1).$$

Differentiate (5). The factor multiplying ax^{p-1} is $p(1+R) + xR'$, which tends to $p > 0$. Thus $f' > 0$ and $f > 0$ on a sufficiently small interval $(0, x_0)$. The endpoint limit is zero, so f maps this interval bijectively onto $(0, f(x_0))$. Finally $y = ag(y)^p(1 + o(1))$, giving $g(y)/t \rightarrow 1$. $\square$

3 The coefficient theorem

Let $D_L = d/dL$. Fix $q > 0$; the desired inverse itself corresponds to $q = 1$. Allowing q also computes powers of the inverse without first making an insufficiently long truncation of g and then raising it to a power.

For $k \neq 0$, define

$$Q_k^{(q)}(L) = \frac{(-1)^{|k|} q}{p^{|k|} k!} \left[\prod_{r=1}^{|k|-1} (q + \Delta_k + pr + D_L) \right] \prod_{j=1}^m P_j(L)^{k_j}. \quad (8)$$

An empty operator product is the identity. Set $Q_0^{(q)} = 1$. All operators in the product commute because their scalar parts are constants with respect to L .

Theorem 3.1 (Constructive positive-real inversion). *For every admissible finite model and every fixed $q > 0$,*

$$g(y)^q = \sum_{k \in \mathbb{N}_0^m} t^{q+\Delta_k} Q_k^{(q)}(\log t) \quad (9)$$

for sufficiently small positive t . The series is absolutely convergent when each polynomial $Q_k^{(q)}$ is regarded as one block. All multiindices at the same weight may therefore be collected. In addition,

$$\deg Q_k^{(q)} \leq \sum_j k_j \deg P_j. \quad (10)$$

Every finite exponent cutoff can be computed by finitely many exact polynomial operations and comparisons of real weights. The resulting expansion is the positive inverse branch specified by $g(y) \sim t$.

The theorem has no rationality requirement on p or the δ_j . For software, exact comparison may require assumptions or may remain undecidable to a particular simplifier; that computational issue does not alter the mathematical statement.

3.1 The first two perturbation depths

Write $P_j = P_j(L)$ and $P'_j = D_L P_j$. Formula (8) yields

$$\begin{aligned} g(y)^q &= t^q - \frac{q}{p} \sum_j t^{q+\delta_j} P_j \\ &\quad + \frac{q}{2p^2} \sum_j t^{q+2\delta_j} ((q+2\delta_j+p)P_j^2 + 2P_j P'_j) \\ &\quad + \frac{q}{p^2} \sum_{i < j} t^{q+\delta_i+\delta_j} ((q+\delta_i+\delta_j+p)P_i P_j + P'_i P_j + P_i P'_j) + \text{higher depths}. \end{aligned} \quad (11)$$

The cross terms are essential. Independently inverting each perturbation and adding the answers would miss them.

3.2 A polynomial-vector implementation

If $Z(L) = \sum_{s=0}^K z_s L^s$, then

$$(A + D_L)Z = \sum_{s=0}^K (Az_s + (s+1)z_{s+1})L^s, \quad z_{K+1} = 0. \quad (12)$$

Thus every operator in (8) acts by a short upper-bidiagonal transformation on the coefficient vector. No unknown coefficient system has to be solved. The supplied Wolfram implementation applies the equivalent polynomial operation `Expand[A z + D[z, e11]]`; a specialized sparse-vector backend is a possible optimization, not a prerequisite for the formula.

4 Proof by homotopy reversion

The essential device is to apply analytic reversion at a *positive* basepoint and expand in an auxiliary parameter. We never pretend that a logarithmic function is analytic at the endpoint zero.

4.1 The homotopy Lagrange–Bürmann identity

Lemma 4.1. *Let H and A be analytic on a disk about $v \neq 0$. For sufficiently small λ , let s_λ be the solution near v of*

$$s_\lambda + \lambda H(s_\lambda) = v.$$

Then

$$A(s_\lambda) = A(v) + \sum_{n \geq 1} \frac{(-\lambda)^n}{n!} \left(\frac{d}{dv} \right)^{n-1} (A'(v)H(v)^n). \quad (13)$$

Proof. Choose a small circle C about v on which $|\lambda H(s)| < |s - v|$. Rouché's theorem gives exactly one zero inside, counted with multiplicity. It is simple, and the argument-principle residue formula gives

$$A(s_\lambda) = \frac{1}{2\pi i} \int_C A(s) \frac{1 + \lambda H'(s)}{s - v + \lambda H(s)} ds.$$

The quotient has the uniformly convergent expansion on C

$$\frac{1 + \lambda H'}{s - v + \lambda H} = \frac{1}{s - v} + \sum_{n \geq 1} \frac{(-1)^{n+1} \lambda^n}{n} \frac{d}{ds} \left(\frac{H(s)^n}{(s - v)^n} \right).$$

Integrate each coefficient by parts around the closed contour. The coefficient of λ^n becomes

$$\frac{(-1)^n}{n} \frac{1}{2\pi i} \int_C \frac{A'(s)H(s)^n}{(s - v)^n} ds.$$

Cauchy's differentiation formula supplies the factor $1/(n-1)!$, proving (13). These are the usual analytic ingredients of Lagrange reversion [2]; here the small parameter is λ , not v . $\square$

For the near-identity case $f(x) = x + h(x)$, choose $A(s) = s$ and $v = y$. The immediately useful formula is

$$g(y) = y + \sum_{n \geq 1} \frac{(-1)^n}{n!} \frac{d^{n-1}}{dy^{n-1}} h(y)^n, \quad (14)$$

provided the homotopy expansion reaches $\lambda = 1$. This proviso is proved below for all admissible finite models.

4.2 Removing the leading power

Set

$$s = x^p, \quad v = y/a = t^p,$$

and define

$$H_j(s) = s^{1+\delta_j/p} P_j \left(\frac{\log s}{p} \right), \quad H(s) = \sum_j H_j(s).$$

The inverse equation becomes $s + H(s) = v$. To obtain $g(y)^q$ use $A(s) = s^{q/p}$ in (13).

Let $n = |k|$, $\Delta = \Delta_k$, and $Z_k(L) = \prod_j P_j(L)^{k_j}$. The multinomial contribution indexed by k in $A'(s)H(s)^n$ is

$$\frac{q}{p} \frac{n!}{k!} s^{n-1+(q+\Delta)/p} Z_k \left(\frac{\log s}{p} \right).$$

The elementary differentiation identity

$$\frac{d^r}{ds^r} \left[s^b Z \left(\frac{\log s}{p} \right) \right] = s^{b-r} \prod_{j=0}^{r-1} \left(b - j + \frac{1}{p} D_L \right) Z(L) \quad (15)$$

follows by induction. Apply it with $r = n - 1$ and $b = n - 1 + (q + \Delta)/p$. Factoring $1/p$ from every operator gives exactly (8), after $s = v = t^p$. This proves the coefficient identity formally and locally in the homotopy parameter. The degree bound follows because D_L only lowers polynomial degree.

4.3 Why the homotopy reaches one and the block series converges

Fix $0 < r < 1$ and consider $|s - v| \leq rv$, where $v = t^p > 0$. A single analytic logarithm exists on this disk. Since s/v remains in the compact disk $|s/v - 1| \leq r$,

$$\frac{\log s}{p} = L + \frac{1}{p} \log(s/v)$$

differs from L by a bounded quantity. For fixed model parameters there are constants $C_j > 0$, independent of sufficiently small t , such that

$$|H_j(s)| \leq C_j v \varepsilon_j(t), \quad \varepsilon_j(t) = t^{\delta_j} T^{K_j}, \quad K_j = \deg P_j. \quad (16)$$

Every $\varepsilon_j(t)$ tends to zero.

Introduce independent parameters λ_j and solve

$$s + \sum_j \lambda_j H_j(s) = v.$$

For $m > 0$, choose the polydisk radii

$$\mathcal{R}_j(t) = \frac{r}{2mC_j \varepsilon_j(t)}.$$

On $|s - v| = rv$ and $|\lambda_j| \leq \mathcal{R}_j$ the sum of the perturbation bounds is at most $rv/2$. Rouché's theorem again gives exactly one simple root inside. Local analytic dependence on the parameters patches to a single analytic root throughout the polydisk because the root is unique. Furthermore $|s/v - 1| < r$, so the observable $s^{q/p}$ is analytic there and bounded in absolute value by Ct^q for a fixed C .

The multivariate Cauchy coefficient estimate now gives

$$\left| t^{q+\Delta_k} \mathcal{Q}_k^{(q)}(L) \right| \leq Ct^q \prod_j \mathcal{R}_j(t)^{-k_j}. \quad (17)$$

All radii tend to infinity. For sufficiently small t , the geometric sum of the right-hand side over $k \in \mathbb{N}_0^m$ is finite. Hence the Taylor series in the λ_j converges absolutely at $(1, \dots, 1)$ and gives (9). Its coefficients agree with the multinomial calculation above, for instance by first substituting $\lambda_j = \lambda \mu_j$ and comparing powers of λ and the μ_j .

At $(1, \dots, 1)$ the root is real by complex-conjugation symmetry and unique; it is positive because it lies in the disk about $v > 0$ of radius less than v . It is the local inverse of [theorem 2.2](#). This completes the proof of [theorem 3.1](#).

Remark 4.2. The convergence assertion groups each polynomial $Q_k^{(q)}(L)$ as one block. It permits reordering and combining equal weights, which is all the algorithm needs. It is not an assertion that an arbitrary ungrouping or resummation convention for more general transseries preserves convergence.

5 Formal uniqueness and finite determination

The analytic proof explains why the result is a genuine inverse. The formal algebra explains why each requested finite part can be computed without examining infinitely many terms.

Consider formal sums

$$\sum_{\Delta \in S} t^\Delta A_\Delta(L), \quad A_\Delta \in \mathbb{R}[L],$$

with S as in [section 2](#). Addition and multiplication are well defined because each bounded weight range contains only finitely many indices. The weight valuation is the smallest exponent with nonzero polynomial coefficient. The Euler derivation acts by

$$t \frac{d}{dt} (t^\Delta A(L)) = t^\Delta (\Delta A + A'),$$

so it preserves weights.

If U has positive valuation, the binomial series $(1 + U)^b$ and the formal series $\log(1 + U)$ are well defined: only finitely many powers of U contribute below any finite weight. Therefore substitution into a power-log term takes the unambiguous form

$$(t(1 + U))^\delta P(\log(t(1 + U))) = t^\delta (1 + U)^\delta P(L + \log(1 + U)). \quad (18)$$

The logarithm addition here is justified analytically by the chosen positive branch and formally by the normalization of $1 + U$; it is not an unrestricted computer-algebra power expansion.

Proposition 5.1 (Formal existence and uniqueness). *There is a unique formal series U with zero constant term such that $g = t(1 + U)$ solves the normalized inverse equation. Each coefficient below a fixed weight depends on only finitely many input coefficients and finite polynomial operations.*

Proof. For $m = 0$, the normalized solution is $U = 0$. For $m > 0$, the equation for U is

$$(1 + U)^p \left(1 + \sum_j t^{\delta_j} (1 + U)^{\delta_j} P_j(L + \log(1 + U)) \right) = 1.$$

Equivalently,

$$U = \left(1 + \sum_j t^{\delta_j} (1 + U)^{\delta_j} P_j(L + \log(1 + U)) \right)^{-1/p} - 1. \quad (19)$$

Start with $U_0 = 0$ and repeatedly apply the right-hand side. A difference of weight w between two inputs produces a difference of weight at least $w + d$ between their outputs, because every perturbation carries a factor of weight at least $d = \min \delta_j > 0$. Thus the iteration stabilizes below any fixed cutoff after finitely many steps and defines a solution.

For uniqueness, suppose two solutions first differ at weight w . In the original normalized equation their difference at weight w is p times that nonzero difference; all perturbation contributions have higher weight. Since $p \neq 0$, both equations cannot vanish. The finiteness assertion follows from local finiteness of the support. $\square$

The independent verification program uses the marker-variable version of (19), not the differential coefficient formula, to check the coefficients. This makes the verification substantially stronger than printing the same formula in two programming languages.

6 Remainders and two different truncation policies

The formulas are most useful when accompanied by a precise interpretation of “order.” The package supports two policies and reports which one was used.

6.1 Truncation by total perturbation depth

For $N \in \mathbb{N}_0$, define

$$G_{q,N}^{\text{depth}}(t) = \sum_{|k| \leq N} t^{q+\Delta_k} \mathcal{Q}_k^{(q)}(L).$$

For $m > 0$ put $d = \min_j \delta_j$ and $K = \max_j \deg P_j$.

Theorem 6.1 (Depth remainder). *For each fixed N ,*

$$g(y)^q - G_{q,N}^{\text{depth}}(t) = O\left(t^{q+(N+1)d} T^{(N+1)K}\right). \quad (20)$$

For $m = 0$ the answer $g(y)^q = t^q$ is exact.

Proof. Apply theorem 4.1 with a single parameter multiplying $H = \sum H_j$. On the disk used in (16),

$$|H(s)| \leq C\nu\varepsilon(t), \quad \varepsilon(t) = t^d T^K \longrightarrow 0.$$

Rouché’s theorem gives a homotopy convergence radius at least $c/\varepsilon(t)$ with a fixed $c > 0$. Cauchy’s estimate bounds the coefficient of depth n by $C't^q (C''\varepsilon(t))^n$. Summing its geometric tail from $n = N + 1$ proves (20). $\square$

With a single gap, depth and exponent cutoffs are closely related. With several gaps they need not agree. For instance, depth one for $f = x + x^2 + x^4$ gives $y - y^2 - y^4$ while the omitted depth-two term $2y^3$ is larger than the retained y^4 term. The depth-one approximation is still valid with an $O(y^3)$ error, but it is not a complete exponent truncation through y^4 . This distinction is explicit in the returned metadata.

6.2 Truncation by an absolute exponent of the output variable

The public exponent cutoff B means: retain every full block whose exponent of y is *strictly less than* B . Since $t = (y/a)^{1/p}$, this is equivalent to retaining

$$\Delta_k < D, \quad D = pB - q.$$

The package requires $B > q/p$, so the leading term is included.

Let

$$I_D = \{k \in \mathbb{N}_0^m : \Delta_k < D\}.$$

For $m > 0$ its finite outer boundary is

$$\partial I_D = \{k + e_j : k \in I_D, k + e_j \notin I_D, 1 \leq j \leq m\}. \quad (21)$$

Define

$$\Delta_* = \min\{\Delta_k : k \in \partial I_D\}, \quad (22)$$

$$K_* = \max \left\{ \sum_j k_j K_j : k \in \partial I_D, \Delta_k = \Delta_* \right\}. \quad (23)$$

These are computed from finitely many exact weights; no common denominator of irrational exponents is introduced.

Lemma 6.2. Δ_* is the smallest weight of any omitted multiindex. Every omitted multiindex of weight Δ_* is on ∂I_D .

Proof. Take a path from 0 to an omitted index by adding coordinate unit vectors. The first point on the path that leaves I_D lies on the boundary, and all subsequent steps strictly increase weight. Thus no omitted index can have weight smaller than the boundary minimum. If an omitted index has this minimum weight, subtracting any occupied coordinate must return to I_D ; otherwise there would be an omitted point of smaller weight. Hence that index itself is a boundary point. $\square$

Theorem 6.3 (Exponent remainder). For $G_q^{<B}(t) = \sum_{k \in I_D} t^{q+\Delta_k} Q_k^{(q)}(L)$,

$$g(y)^q - G_q^{<B}(t) = O\left(t^{q+\Delta_*} T^{K_*}\right). \quad (24)$$

The bound remains valid when different contributions at Δ_* cancel; in that event it can overestimate the actual remainder.

Proof. Choose an integer N with $(N+1)d > \Delta_*$. By [theorem 6.1](#), the infinite tail after depth N is

$$O(t^{q+(N+1)d} T^{(N+1)K}) = o(t^{q+\Delta_*}).$$

There are finitely many remaining omitted indices with depth at most N . Those of weight Δ_* have polynomial degree at most K_* by [theorem 6.2](#) and (10). Every one with strictly larger weight is $o(t^{q+\Delta_*})$ by (7). This proves the claimed bound. Cancellation only reduces the finite leading contribution. $\square$

If the cutoff falls in a gap of the support, $\Delta_* > D$ and the actual reported power is sharper than B . At a support point the logarithmic degree must be retained: (24) is not automatically $O(y^B)$. The package's remainder is an *asymptotic scale*, not an explicit numerical constant or a computed interval of validity.

7 The logarithmic example to arbitrary order

For $f(x) = x + x^2(1 + \log x)$, take $a = p = 1$, $m = 1$, $\delta_1 = 1$, and $P_1(L) = 1 + L$. Formula (8) becomes

$$Q_n(L) = \frac{(-1)^n}{n!} \prod_{r=1}^{n-1} (n+1+r+D_L)(1+L)^n, \quad Q_0 = 1, \quad (25)$$

and

$$g(y) = \sum_{n \geq 0} y^{n+1} Q_n(\log y). \quad (26)$$

The first coefficient polynomials, generated independently by the verification program, are

$$Q_0(L) = 1 \quad (27)$$

$$Q_1(L) = -L - 1 \quad (28)$$

$$Q_2(L) = 2L^2 + 5L + 3 \quad (29)$$

$$Q_3(L) = -5L^3 - \frac{41L^2}{2} - 27L - \frac{23}{2} \quad (30)$$

$$Q_4(L) = 14L^4 + \frac{241L^3}{3} + \frac{335L^2}{2} + 151L + \frac{299}{6} \quad (31)$$

For example the third correction can also be checked directly:

$$-\frac{1}{6} \frac{d^2}{dy^2} (y^6 (1 + \log y)^3) = -y^4 \left(5(1 + \log y)^3 + \frac{11}{2} (1 + \log y)^2 + (1 + \log y) \right).$$

Expansion of this expression gives the fourth-power block in (2).

7.1 Catalan numbers and harmonic numbers in the coefficients

The highest logarithmic degree in Q_n is exactly n . Its coefficient is

$$[L^n]Q_n(L) = \frac{(-1)^n}{n!} \prod_{s=n+2}^{2n} s = (-1)^n \frac{1}{n+1} \binom{2n}{n} = (-1)^n C_n. \quad (32)$$

The empty product convention covers $n = 1$. This identity supplies an inexpensive high-order consistency check.

Let $H_j = \sum_{s=1}^j 1/s$. The next coefficient is

$$[L^{n-1}]Q_n(L) = (-1)^n C_n n(1 + H_{2n} - H_{n+1}), \quad n \geq 1. \quad (33)$$

To prove it, the constant part of the operator product contributes the coefficient n of L^{n-1} in $(1 + L)^n$, while choosing one derivative contributes an additional $n \sum_{s=n+2}^{2n} 1/s$. Two or more derivatives cannot contribute at this degree. For $n = 3$ the formula gives $-41/2$, agreeing with the explicit block.

More generally let e_j denote an elementary symmetric polynomial, with $e_0 = 1$. Then

$$Q_n(L) = \frac{(-1)^n}{n!} \sum_{j=0}^{n-1} e_{n-1-j}(n+2, n+3, \dots, 2n) \frac{n!}{(n-j)!} (1 + L)^{n-j}. \quad (34)$$

This is a finite nonrecursive specification of every coefficient. It follows by expanding the commuting operator product and differentiating $(1 + L)^n$.

7.2 The correct remainder after a displayed truncation

For each fixed N ,

$$g(y) = \sum_{n=0}^N y^{n+1} Q_n(\log y) + O(y^{N+2} |\log y|^{N+1}). \quad (35)$$

In particular,

$$g(y) = y - y^2(1 + \log y) + O(y^3 |\log y|^2).$$

Replacing this last remainder by $O(y^3)$ would be false, since the next block has a nonzero $2 \log^2 y$ coefficient. This corrects the informal order notation used in the original discussion without changing its leading two terms.

7.3 An explicit sufficient convergence condition and tail bound

In this example the analytic argument can be made quantitative without solving for a critical point. On the disk $|z - y| \leq y/2$, $0 < y < 1$,

$$|z| \leq \frac{3}{2}y, \quad |\log(z/y)| \leq \sum_{j \geq 1} \frac{(1/2)^j}{j} = \log 2.$$

Thus, with $A(y) = 1 + |\log y| + \log 2$,

$$|h(z)| = |z^2(1 + \log z)| \leq \frac{9}{4}y^2 A(y).$$

Put $\rho(y) = 9yA(y)$ and choose a homotopy radius $\mathcal{R} = 1/\rho(y)$. On $|\lambda| \leq \mathcal{R}$, $|\lambda h(z)| \leq y/4 < y/2$. The inverse root satisfies $|g_\lambda(y) - y| < y/2$. Cauchy's estimate applied to $g_\lambda - y$ gives the explicit bound

$$\left| g(y) - \sum_{n=0}^N y^{n+1} Q_n(\log y) \right| \leq \frac{y \rho(y)^{N+1}}{2(1 - \rho(y))}, \quad \rho(y) < 1. \quad (36)$$

This is sufficient, not optimal. It proves ordinary convergence of the homotopy block sum for every positive y satisfying the condition and yields a computable error bound there. The sharper asymptotic order in (35) follows immediately, although its best constant need not resemble the conservative constant in (36).

8 The irrational-power example and its convergence radius

Let $\alpha > 1$, $d = \alpha - 1$, and $f_\alpha(x) = x + x^\alpha$. Then

$$g_\alpha(y) = \sum_{n \geq 0} c_n y^{1+nd}, \quad c_0 = 1, \quad (37)$$

where for $n \geq 1$

$$c_n = \frac{(-1)^n}{n!} \prod_{r=0}^{n-2} (n\alpha - r) \quad (38)$$

$$= \frac{(-1)^n \Gamma(n\alpha + 1)}{\Gamma(n+1) \Gamma(nd + 2)} = \frac{(-1)^n}{1 + nd} \binom{n\alpha}{n}. \quad (39)$$

Formula (38) follows directly from $(-1)^n D_y^{n-1} y^{n\alpha}/n!$. It is preferable to a gamma quotient in symbolic software because it is a finite polynomial in α for each n and has no apparent special-function singularities to simplify.

The first coefficients are

$$\begin{aligned} c_1 &= -1, & c_2 &= \alpha, \\ c_3 &= -\frac{\alpha(3\alpha - 1)}{2}, & c_4 &= \frac{\alpha(4\alpha - 1)(4\alpha - 2)}{6}. \end{aligned}$$

For $\alpha = \sqrt{2}$ these give (3); the next two are

$$c_5 = -\frac{305}{12} + \frac{51\sqrt{2}}{4}, \quad c_6 = -77 + \frac{322\sqrt{2}}{5}.$$

The powers are exactly $1 + n(\sqrt{2} - 1)$. Replacing $\sqrt{2}$ by a decimal number would change the problem and would destroy exact identities between weights in more complicated examples.

8.1 A regular power series in the correct auxiliary variable

Set $z = y^d$ and $u = g_\alpha(y)/y$. The equation is

$$u + zu^\alpha = 1, \quad (40)$$

so $u(z) = \sum_{n \geq 0} c_n z^n$ is an ordinary analytic power series near $z = 0$. This is the right way to reduce the irrational-power problem to an analytic implicit equation. Trying to force the original expansion into integer powers of y loses the natural scale.

Proposition 8.1. *The radius of convergence of $u(z)$ is*

$$R_\alpha = \frac{(\alpha - 1)^{\alpha-1}}{\alpha^\alpha}. \quad (41)$$

The series (37) therefore converges for $0 < y < R_\alpha^{1/(\alpha-1)}$. Its block coefficients satisfy

$$|c_n| \sim \frac{\sqrt{\alpha}}{\sqrt{2\pi}(\alpha - 1)^{3/2}} R_\alpha^{-n} n^{-3/2}. \quad (42)$$

Proof. Apply Stirling's formula to the gamma quotient in (39), with $d = \alpha - 1 > 0$. The numerator is asymptotic to $\sqrt{2\pi\alpha n}(\alpha n/e)^{\alpha n}$, while the denominator is asymptotic to $\sqrt{2\pi n}(n/e)^n \sqrt{2\pi}(dn)^{dn+3/2} e^{-dn}$. Cancellation gives (42). The Cauchy–Hadamard root test then gives (41) exactly. $\square$

The associated critical point is also visible directly. Differentiating the left side of (40) with respect to u and setting it to zero gives $1 + \alpha zu^{\alpha-1} = 0$. Together with the equation this yields $u = \alpha/(\alpha - 1)$ and $z = -R_\alpha$. This identifies the negative-real singularity limiting the Taylor series in z , even though the positive real inverse exists for all $y > 0$.

For fixed N , the next exponent is $1 + (N + 1)d$, so the finite-order estimate is particularly simple:

$$g_\alpha(y) = \sum_{n=0}^N c_n y^{1+nd} + O(y^{1+(N+1)d}).$$

There are no logarithmic factors to include in this example.

9 Mixed scales, collisions, and nonunit leading powers

9.1 Two independent power corrections

For

$$f(x) = x + Ax^\alpha + Bx^\beta, \quad \alpha, \beta > 1,$$

put $d_1 = \alpha - 1$ and $d_2 = \beta - 1$. The depth-two expansion is

$$g(y) = y - Ay^\alpha - By^\beta + A^2\alpha y^{2\alpha-1} + AB(\alpha + \beta)y^{\alpha+\beta-1} + B^2\beta y^{2\beta-1} + O(y^{1+3\min(d_1, d_2)}). \quad (43)$$

The error describes depth truncation, so some of the displayed higher-power terms can be smaller than the error. For a complete ordered expansion one instead chooses an exponent cutoff and includes every $k_1 d_1 + k_2 d_2$ below it.

For $n = k_1 + k_2 \geq 1$, the coefficient at the labelled multiindex is

$$\frac{(-1)^n A^{k_1} B^{k_2}}{k_1! k_2!} \prod_{r=0}^{n-2} (k_1 \alpha + k_2 \beta - r),$$

multiplying $y^{1+k_1 d_1 + k_2 d_2}$. This is the mixed generalized-binomial analogue of (38). Polynomial logarithmic coefficients are handled by replacing each scalar factor with the corresponding operator in (8).

9.2 An exact cancellation caused by a collision

For $f = x + x^2 + x^3$, the gaps are 1 and 2. The inverse starts

$$g(y) = y - y^2 + y^3 - 4y^5 + 14y^6 - 30y^7 + 33y^8 + O(y^9). \quad (44)$$

There is no y^4 term. At weight 3, the multiindex $(3, 0)$ contributes -5 and $(1, 1)$ contributes $+5$. Both must be computed and added. A coefficient engine that assigns one term to each input generator without cross products, or that overwrites an earlier term at the same weight, fails this test.

The remainder estimator deliberately uses a support candidate before relying on cancellation. It remains correct when the candidate coefficient vanishes; it may then be conservative. Automatically seeking the first nonzero coefficient beyond an arbitrary number of cancellations is not needed to satisfy a finite requested cutoff.

9.3 A ramified positive inverse

For

$$f(x) = 3x^2(1 + x(1 + \log x)), \quad t = \sqrt{y/3}, \quad P = 1 + \log t,$$

the first terms are

$$g(y) = t - \frac{1}{2}t^2P + t^3\left(\frac{5}{8}P^2 + \frac{1}{4}P\right) - \frac{t^4}{48}(6 + D_L)(8 + D_L)P^3 + O(t^5T^4). \quad (45)$$

The leading inverse is a square root, but the positive normalization makes the branch unambiguous. The cutoff must still refer to the exponent of y , not of t : retaining terms with y exponent below $5/2$ retains the displayed powers through t^4 .

As an independent algebraic check, for $f(x) = 3x^2(1 + x^2)$ the exact positive inverse is

$$g(y) = \sqrt{\frac{-1 + \sqrt{1 + 4y/3}}{2}} = t\sqrt{\frac{2}{1 + \sqrt{1 + 4t^2}}}.$$

Expansion of the last ordinary analytic factor agrees with (8). The verification program checks this identity through the available finite order without using the same differential formula twice.

9.4 Powers of the inverse

For $f = x + x^2$, the inverse itself is $(\sqrt{1 + 4y} - 1)/2$. Setting $q = 2$ in (8) gives directly

$$g(y)^2 = y^2 - 2y^3 + 5y^4 - 14y^5 + O(y^6).$$

This is a simple test of the factor $q/p^{|k|}$ in the general formula. For arbitrary p , omitting that normalization factor would produce incorrect coefficients even if all near-identity examples happened to pass.

10 Residual certificates, forward remainders, and Newton refinement

10.1 From a forward residual to an inverse error

Proposition 10.1 (Residual transport). *Suppose g is the real inverse of f , G is a positive approximation, and $f'(x) \geq m > 0$ on the interval between G and $g(y)$. Then*

$$|G - g(y)| \leq \frac{|f(G) - y|}{m}. \quad (46)$$

Proof. Apply the mean value theorem to $f(G) - f(g(y))$. □

For the logarithmic example, the global constant m_0 in (4) is valid for every positive approximation. For $x + x^\alpha$, one may use $m = 1$. These are rigorous analytic inequalities. Evaluating either side with ordinary arbitrary-precision floating-point arithmetic is still not the same thing as interval certification.

For the general model, an approximation with $G/t \rightarrow 1$ and the true inverse both lie in $[t/2, 2t]$ eventually. From [theorem 2.2](#), eventually

$$f'(x) \geq ap 2^{-1-|p-1|} t^{p-1} \quad (t/2 \leq x \leq 2t).$$

Thus

$$|G - g(y)| \leq \frac{2^{1+|p-1|}}{ap} t^{1-p} |f(G) - y| \quad (47)$$

for sufficiently small t . In particular, an inverse error of order $t^{1+\sigma} T^K$ corresponds to a forward residual of order $t^{p+\sigma} T^K$, not generally the same absolute exponent. This scaling is important when $p \neq 1$.

10.2 Using a finite forward expansion of a more complicated function

The package parses a finite expression exactly. A separate approximation to a more complicated forward function may nevertheless be used with a justified error budget.

Proposition 10.2 (Transport of an omitted forward tail). *Let f_0 be an admissible finite model and suppose an actual real function $\tilde{f}$ satisfies, near zero,*

$$\tilde{f}(x) - f_0(x) = O(x^{p+\sigma} (1 + |\log x|)^K), \quad \sigma > 0.$$

Assume also that $\tilde{f}'(x) \sim apx^{p-1}$ and that its positive inverse $\tilde{g}$ exists. Then

$$\tilde{g}(y) - g_0(y) = O(t^{1+\sigma} T^K). \quad (48)$$

Proof. The hypotheses give $\tilde{g}(y) \sim t$ and $g_0(y) \sim t$. Evaluate the forward discrepancy at $g_0(y)$, obtaining a residual of order $t^{p+\sigma} T^K$. Apply the scaled slope estimate to $\tilde{f}$ on the interval between its inverse and g_0 . □

The derivative hypothesis must not be silently inferred by differentiating an arbitrary Big-O estimate. Without derivative control, a tiny oscillatory forward error can have a large derivative and can even spoil local monotonicity. For an explicitly controlled analytic expansion such derivative bounds may be available, but they are part of the input justification.

This proposition gives a principled way to preprocess, for example, an analytic trigonometric factor by a sufficiently long Taylor polynomial. The inverse's reported algorithmic remainder must then be combined with (48). Increasing the inverse cutoff beyond the accuracy of the supplied forward model does not create additional information about the actual function.

10.3 One Newton correction can improve an approximation

Let $G \sim t$, set $r = f(G) - y$, and define

$$G_+ = G - \frac{r}{f'(G)}.$$

For admissible finite models, $f''(x) = O(t^{p-2})$ on $[t/2, 2t]$ and $f'(G) \asymp t^{p-1}$. Taylor's theorem therefore gives

$$G_+ - g(y) = O\left(\frac{(G - g(y))^2}{t}\right).$$

Hence an error $O(t^{1+\sigma}T^K)$ improves to $O(t^{1+2\sigma}T^{2K})$, as long as the points remain in the local interval. This is an analytic or numerical refinement; a symbolic Newton expression must still be truncated in the correct scale to be called a canonical power–log expansion.

11 Algorithm, representation, and computational correctness

11.1 Normalization and the conservative parser

The automatic parser first replaces the literal subexpression $\text{Log}[x]$ by a fresh symbol L and expands the finite sum. Each summand must be a product of explicit powers x^β and factors independent of x ; the remaining coefficient must be a polynomial in L . Equal exponents that can be proved equal are combined.

Next it identifies a provably least input exponent p . Its coefficient must be a positive constant a , not a nonconstant polynomial in L . Every other term is divided by ax^p , producing positive gaps δ_j and polynomials P_j . Realness and positivity are checked under the supplied assumptions. The parser does not have to decide an ordering between every pair of higher symbolic input exponents merely to recognize a least leading term.

The grammar is intentionally narrower than mathematical equivalence under all possible branch conditions. For example, the expression $\text{Log}[x^2]$ is not automatically rewritten as $2 \text{Log}[x]$. A user who has already proved the appropriate positive-domain identity can supply the normalized expression or an explicit model. This keeps branch-dependent simplification out of the parsing layer.

11.2 Enumerating exactly the required multiindices

Exponent-mode enumeration starts at $k = 0$ and explores children $k + e_j$. A child with weight below $D = pB - q$ is accepted and expanded in turn. A child at or above the cutoff is recorded on the finite boundary but is not expanded. A dictionary of integer multiindices avoids duplicate visits.

Every accepted index is reached because all its predecessors have smaller weight. Every boundary candidate is reached from its accepted parent. Conversely, no omitted subtree can re-enter the accepted region because all gaps are positive. These statements prove the enumeration step independently of any particular data structure.

For depth mode, the same traversal uses $|k| \leq N$ instead of a weight inequality. Its output is finite without ordering the symbolic gaps. The number of accepted indices is $\binom{N+m}{m}$ before any normalizing cancellations. Exponent-mode finiteness also follows from $|k| < D/d$; a crude enclosing box has at most $\prod_j (1 + \lfloor D/\delta_j \rfloor)$ grid points.

11.3 Exact weights and equality policy

The implementation asks `FullSimplify` to prove equality, strict less than, or strict greater than under the supplied assumptions. There is no floating-point fallback. If an exponent-mode comparison cannot be resolved, the function returns `Failure["UndecidableOrder", ...]` rather than silently sorting approximate values.

In depth mode, a relative order may be unnecessary. Proven equalities are collected, while unresolved possible coincidences remain an additive sum. This representation is still algebraically valid; it does not claim the weights are distinct. The metadata states that the ordering is by homotopy depth rather than a canonical increasing-exponent list.

Machine and arbitrary-precision `Real` inputs are rejected, including inexact coefficients as well as exponents. This rule is deliberately strict: an exact identity between exponents is part of the mathematical input, not a numerical tolerance decision. A user may choose an exact rational model explicitly, but the package does not choose one on the user's behalf.

11.4 Coefficient evaluation, collection, and reconstruction

For each included index, compute Δ_k , the polynomial product $\prod P_j^{k_j}$, and the operator sequence in (8). Contributions with provably equal weights are added before output. Proven zero polynomials are removed. The reconstructed expression is

$$\sum_k \left(\frac{y}{a}\right)^{(q+\Delta_k)/p} Q_k^{(q)}\left(\frac{\log(y/a)}{p}\right). \quad (49)$$

This follows from the positive-domain normalization directly; the code does not use `PowerExpand` to transform nested powers.

The result deliberately is not an ordinary `SeriesData` object. `SeriesData` represents a fixed-denominator sequence of exponents, and `InverseSeries` is documented as inversion of a power series [6, 5]. Such a representation is not a universal container for finite combinations of arbitrary irrational generators. Moreover, a logarithmic error bound deserves explicit metadata even where some built-in series operations can manipulate logarithmic coefficients. The package does not redefine any built-in asymptotic function.

11.5 Resource limits and the actual implementation cost

The option "MaxTerms" bounds the number of visited labelled multiindices, including boundary candidates. Exceeding it returns a failure; no partial result is returned with an apparently complete remainder.

This is not a bound on all symbolic work. The current collector scans its existing blocks for exact equality, so it can perform quadratically many simplification queries in the number of blocks. Polynomial expression swell and difficult parameter inequalities may also dominate execution time. The source is intended as a transparent reference-quality implementation for moderate finite orders, not as a claim of optimal high-order performance. Canonical algebraic-number hashing, cached polynomial products, and a sparse-vector implementation of (12) are natural future optimizations.

Proposition 11.1 (Conditional computational specification). *If the implementation's exact real comparisons and polynomial operations return their mathematically correct values, normalization accepts precisely its stated grammar under the proved conditions, enumeration finishes within its resource bound, and the output is constructed as described, then the returned expression and remainder metadata satisfy theorems 3.1, 6.1 and 6.3.*

Proof. Normalization gives theorem 2.1. The traversal argument gives exactly the required finite index set and its boundary. The coefficient routine implements (8), and collection preserves the sum. Reconstruction is (49). The cited remainder theorems apply to the recorded truncation policy. $\square$

This proposition is a mathematical specification and algorithmic correctness argument, not a machine-checked proof of the Wolfram evaluator or an assertion that native runtime testing has occurred.

12 Wolfram Language package reference

The package context is `PowerLogInverse``. Load the file from its unpacked location; no installation, network request, or repository write is performed by the package.

```
Get["../power-log-inverse/Kernel/PowerLogInverse.wl"];
Clear[x, y, ell];
```

Use distinct, unassigned symbols for the input variable, output variable, and explicit logarithm variable.

12.1 The main expression and data interfaces

```
PowerLogInverse[f, {x, y, B}, options]
PowerLogInverseData[f, {x, y, B}, options]
```

The first returns the explicit inverse approximation. The second returns an association containing that expression, its normalization, branch conditions, coefficient blocks, enumeration details, and remainder information. The cutoff B is a strict exponent of y , not a number of printed monomials.

The options are:

Option	Default	Meaning
Assumptions	True	Fixed-parameter conditions used to prove realness, positivity and exact comparisons.
"Truncation"	"Exponent"	Either strict output-exponent cutoff or "Depth", which interprets the last argument as a nonnegative integer depth.
"InversePower"	1	A fixed positive q ; computes $g(y)^q$.
"MaxTerms"	10000	Maximum visited multiindices, including the first rejected boundary.

The key fields of a successful result are:

Field	Interpretation
"Expression"	The finite expression in the output variable.
"NormalizedExpression"	The same finite sum in fresh symbols t, L , before applying the positive normalization.
"NormalizationRules"	Rules $t \mapsto (y/a)^{1/p}$ and $L \mapsto \log(y/a)/p$.
"Model"	Validated leading coefficient, power, perturbations, logarithm variable, and assumptions.
"Terms"	List of weights, normalized exponents, exponents of y , and logarithmic polynomials.
"Remainder"	Either exact zero, or an association with "Scale", "YExponent", "LogDegree", and the candidate omitted weight.
"Conditions"	Supplied model assumptions together with $y > 0$.
"Branch"	"PositiveNearZero".
"IncludedMultiindices"	Labelled indices used in the finite coefficient sum.
"VisitedCount"	Count used by the enumeration resource guard.
"Ordering"	Increasing exponent or a depth-mode qualification.

A remainder field of type "Big0" means that the unknown error is bounded by a constant times "Scale" for sufficiently small positive y , with fixed admissible parameters. It does not identify the constant or the starting point of that asymptotic regime. The main expression interface omits the metadata for convenience; its mathematical domain remains the same.

12.2 Explicit models and a single coefficient

```
m = PowerLogModel[a, p, {{delta1, P1}, {delta2, P2}}, ell,
  Assumptions -> conditions];
PowerLogInverse[m, {y, B}]
PowerLogInverseData[m, {y, B}]
PowerLogCoefficient[m, {k1, k2}]
```

Here P_1, P_2 are polynomials in the symbol `ell`. The model specifies (5); it is not evaluated at a numerical x . The coefficient function returns exactly (8), as a polynomial in the model's logarithm variable. It also accepts `Assumptions` and `"InversePower"`.

```
m = PowerLogModel[1, 1, {{1, 1 + ell}}, ell];
PowerLogCoefficient[m, {6}]
```

This asks for the coefficient of y^7 in the logarithmic example. Automatic normalization is also available separately as `PowerLogModel[f, x]`. If repeated gaps combine or zero perturbations disappear, the length of the multiindex must match the *normalized* perturbation list stored in the model, not an obsolete unnormalized input list.

12.3 Descriptive failures

Unsupported input returns a `Failure` rather than an arbitrary branch or an unevaluated expression that looks like an answer. Important tags include `"InexactInput"`, `"LogarithmicLeadingTerm"`, `"NonPolynomialLog"`, `"UnprovedPositive"`, `"UnprovedReal"`, `"UndecidableLeadingTerm"`, `"UndecidableOrder"`, `"CutoffTooLow"`, `"InvalidDepth"`, `"VariableCollision"`, and `"TermLimit"`.

For instance,

```
PowerLogInverse[x Log[x], {x, y, 3}]
PowerLogInverse[x + x^1.4, {x, y, 3}]
PowerLogInverse[x + x^2/Log[x], {x, y, 3}]
```

are intentionally rejected for different reasons. The first needs a logarithmic leading core, the second supplies an inexact exponent, and the third has a nonpolynomial logarithmic coefficient. No claim is made that the underlying mathematical problems are impossible; they are outside this package's stated finite polynomial-logarithm grammar.

13 Worked package calls

13.1 The exact original use case

```
d = PowerLogInverseData[x + x^2 (1 + Log[x]), {x, y, 5}];
Collect[d["Expression"], y]
d["Remainder"]
```

The expression is (2) without its order term. The remainder metadata has output exponent 5 and logarithmic degree 4, hence scale $y^5(1 + |\log y|)^4$.

For only the two leading blocks, replace 5 by 3. For arbitrary depth N , use an exact integer N and the depth option:

```
PowerLogInverse[x + x^2 (1 + Log[x]), {x, y, 8},
  "Truncation" -> "Depth"]
```

This includes depths 0 through 8, or powers y through y^9 , not just eight individual logarithmic monomials.

13.2 The irrational-power use case

```
PowerLogInverse[x + x^Sqrt[2], {x, y, 4 Sqrt[2] - 3}]
```

The cutoff is exactly the exponent of the first omitted term in the archive, so the output contains its four requested terms. A generic $\alpha > 1$ is especially convenient in depth mode:

```
PowerLogInverse[x + x^alpha, {x, y, 5},
  "Truncation" -> "Depth", Assumptions -> alpha > 1]
```

An exponent cutoff for symbolic α can require inequalities involving integer multiples of $\alpha - 1$; the appropriate number of included terms can change with α . The package does not silently choose one of those parameter regions.

13.3 Mixed exact irrational gaps

```
mixed = PowerLogInverseData[
  x + x^Sqrt[2] (1 + Log[x]) + 2 x^Sqrt[3],
  {x, y, 5/2}];
mixed["Terms"]
mixed["Remainder"]
```

Every weight generated by $\sqrt{2} - 1$ and $\sqrt{3} - 1$ below $3/2$ is included. The logarithmic polynomial at each weight is produced by the same finite operator formula. There is no conversion to a rational Puiseux denominator.

13.4 Checking an approximation by a forward residual

For the first example a numerical check can be performed as follows:

```
value = 10^-6;
approx = N[d["Expression"] /. y -> value, 80];
residual = approx + approx^2 (1 + Log[approx]) - value;
N[Abs[residual]/(1 - 2 Exp[-5/2]), 30]
```

The expression before numerical evaluation is an upper bound on the true absolute inverse error by [theorem 10.1](#). Its floating-point evaluation is not an outward-rounded interval. A rigorous numerical certificate would also enclose rounding errors in the evaluation of the residual and slope.

14 Executed verification and reproducibility

14.1 What was actually checked

The independent program `Verification/verify.py` ran successfully with Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0. It records 155 successful checks in `verification-results.json` and `verification-report.txt`.

The exact checks include direct differentiation of the logarithmic homotopy formula through depth ten; the falling-factorial formula for arbitrary α through depth ten; the Catalan leading-log coefficient; formal Picard inversion in independent marker variables; substitution into the normalized forward equation; degree bounds; a ramified exact radical inverse; coincident weights and their cancellation; and

comparison of weighted breadth-first enumeration with an independent bounded enumeration.

The formal-jet tests include nonunit leading power, rational and irrational gaps, two logarithmic generators, and a positive noninteger power of the inverse. Each jet operation uses ordinary finite polynomial multiplication, truncated binomial powers, and the Taylor series of $\log(1 + U)$. It does not use the theorem being tested to construct the independent Picard solution.

14.2 Numerical comparisons

For each model, high-precision bisection on a positive bracket supplies a reference inverse. The table uses 100 decimal digits and reports the actual absolute error and the evaluated residual divided by the global slope lower bound. The integer N is the last retained perturbation depth; the sum includes the leading depth-zero term. All entries are computed results, not fitted error laws.

Table 2: Independent numerical checks; floating point, not interval certification.

Model	y	N	Absolute inverse error	Residual/slope bound
log	10^{-2}	1	2.42021×10^{-5}	2.71125×10^{-5}
log	10^{-2}	3	1.48666×10^{-7}	1.66535×10^{-7}
log	10^{-2}	5	1.16329×10^{-9}	1.30311×10^{-9}
log	10^{-4}	1	1.26850×10^{-10}	1.51531×10^{-10}
log	10^{-4}	3	5.09642×10^{-16}	6.08803×10^{-16}
log	10^{-4}	5	2.71946×10^{-21}	3.24859×10^{-21}
log	10^{-8}	1	5.89540×10^{-22}	7.05334×10^{-22}
log	10^{-8}	3	1.16393×10^{-34}	1.39254×10^{-34}
log	10^{-8}	5	3.07789×10^{-47}	3.68243×10^{-47}
sqrt2	10^{-2}	1	2.51975×10^{-4}	3.01769×10^{-4}
sqrt2	10^{-2}	3	1.53364×10^{-5}	1.83843×10^{-5}
sqrt2	10^{-2}	5	1.16850×10^{-6}	1.40080×10^{-6}
sqrt2	10^{-4}	1	6.63135×10^{-8}	6.83614×10^{-8}
sqrt2	10^{-4}	3	9.09723×10^{-11}	9.37821×10^{-11}
sqrt2	10^{-4}	5	1.54532×10^{-13}	1.59305×10^{-13}
sqrt2	10^{-8}	1	3.33239×10^{-15}	3.33468×10^{-15}
sqrt2	10^{-8}	3	2.23019×10^{-21}	2.23172×10^{-21}
sqrt2	10^{-8}	5	1.84415×10^{-27}	1.84541×10^{-27}

The label `log` denotes (1), and `sqrt2` denotes $x + x^{\sqrt{2}}$. For example, at $y = 10^{-8}$ and depth five the errors are approximately $3.07789 \cdot 10^{-47}$ and $1.84415 \cdot 10^{-27}$, respectively. The much slower improvement in the second case is consistent with its small gap $\sqrt{2} - 1$ rather than a unit gap.

The JSON report also compares the signed error with the first omitted block. For the logarithmic case at depth five and $y = 10^{-8}$ that ratio is about 1.0000005439; for the irrational-power case it is about 0.9990508345. These observations support the leading-remainder analysis but do not replace the proofs.

14.3 What was not executed

The connected Wolfram context/evaluator service returned HTTP 404, and no local Wolfram kernel was available. Therefore the package's native runtime behavior and the 35 supplied MUnit regression tests were *not* executed in this session. The source received a manual audit and a lexical string/comment/delimiter check. That lexical checker is not a Wolfram parser and is not evidence of runtime correctness.

The native regression suite covers the original examples, ordinary `InverseSeries` cross-checks, explicit models, symbolic assumptions, normalization, observables, collisions, remainder metadata, and expected failures. Run it in a native installation before integrating the package into a larger workflow:

```
wolframscript -file Tests/RunTests.wl
```

Alternatively evaluate `TestReport[".../Tests/PowerLogInverse.wlt"]` in a notebook. The supplied command-line runner records the kernel version and native report. It supports the current documented `"ReportSucceeded"` property and checks available properties before using the legacy success counts [8, 9, 10]. No fabricated native test report is included.

14.4 Rebuilding the article and the verification outputs

From the unpacked archive root, run

```
python Verification/verify.py
python Verification/static_check.py
cd Article
pdflatex -interaction=nonstopmode -halt-on-error power-log-inverse.tex
pdflatex -interaction=nonstopmode -halt-on-error power-log-inverse.tex
```

The verification program regenerates the numerical table and coefficient blocks included by the article. The document uses standard TeX Live packages and an inline bibliography; it needs no network access or bibliography processor. The software license and article license are recorded separately in the release.

15 Limits of the implementation and principled extensions

15.1 A different leading core is a different problem

A model such as $x \log(1/x)$ has a logarithmic leading factor, so it is not of [theorem 2.1](#). Its positive inverse near zero uses the large negative Lambert branch: writing $u = \log(1/x)$ gives $y = ue^{-u}$ and hence

$$x = -\frac{y}{W_{-1}(-y)}.$$

One can then study perturbations around this exact logarithmic core. The normalization is different from $t = (y/a)^{1/p}$ and requires a different coefficient algebra. Rejecting the input in the present package is preferable to disguising that change of problem as a power-series inversion.

Likewise, negative or fractional powers of $-\log x$, iterated logarithms, and oscillatory coefficients can require a richer coefficient ring and new error estimates. The polynomial logarithm condition is what makes every coefficient here a finite, explicitly representable polynomial.

15.2 The homotopy identity is more general than the parser

The analytic identity (13) applies whenever its local analyticity and smallness hypotheses hold. Some exponentially small perturbations or other analytic germs on punctured domains satisfy those hypotheses even though they are not finite power-log polynomials. The universal differential formula can therefore be a starting point for further work.

What does not follow automatically is a useful ordering of the resulting scales, a finite support below a requested cutoff, or the logarithmic degree bound used by this implementation. Those are separate structural properties, and this release does not advertise an unrestricted logarithmic-exponential transseries engine.

15.3 Fixed parameters versus uniform asymptotics

All constants in the convergence and remainder proofs may depend on p , the gaps, coefficients, and polynomial degrees. The estimates are not uniform as $p \rightarrow 0$, a gap tends to zero, or the number of generators grows. In particular, when $\alpha \downarrow 1$, the natural gap in the irrational-power example shrinks and a fixed exponent cutoff contains more and more depths.

The proof explains the obstruction rather than hiding it: the local-finiteness bound uses $d = \min \delta_j > 0$, and the homotopy coefficient estimate uses $t^d T^K \rightarrow 0$ for fixed d . A simultaneous limit may require a new rescaled leading equation.

15.4 Useful next implementation steps

The most valuable extensions are concrete: a native-kernel compatibility run and continuous regression testing; cached sparse polynomial vectors; canonical exact-algebraic weight keys; optional interval evaluation of residual certificates; and a forward-expansion object carrying both function and derivative remainder contracts as in [theorem 10.2](#). Extending to nonpolynomial logarithmic coefficients should come with a corresponding remainder theorem rather than only an additional parser rewrite.

16 Conclusion

The requested positive inverse expansions are obtained by choosing the correct endpoint germ and reverting a small perturbation of an explicit monomial core. Logarithms and irrational powers then cease to be exceptional cases: they are coefficients and weights in a locally finite algebra.

The central formula (8) computes every requested coefficient by finite polynomial operations. The homotopy argument proves that the formal blocks represent the genuine positive inverse near zero, and the boundary construction supplies valid logarithm-aware remainder metadata for exact exponent cutoffs. The source package implements this specification without silently rationalizing powers or choosing an arbitrary complex branch.

The archive includes the complete implementation, examples, native regression tests, and the executed independent mathematical checks. The remaining software verification limitation is explicit: the native Wolfram tests still need to be run in an available kernel.

A A minimal near-identity reference implementation

For the two original examples, both of which have $f(x) = x + h(x)$ with $h = o(x)$, the essential finite formula fits in a few lines. This educational routine assumes the analytic hypotheses; it does not replace the validated parser, weight collector, or remainder metadata of the main package.

```
Clear[NearIdentityInverse];
NearIdentityInverse[f_, {x_Symbol, y_Symbol}, n_Integer] /;
  (n >= 0 && x != y) :=
Module[{h = f - x},
  Expand[y + Total[Table[
    ((-1)^k/Factorial[k] D[h^k, {x, k - 1}]) /. x -> y,
    {k, 1, n}]]]
];

NearIdentityInverse[x + x^2 (1 + Log[x]), {x, y}, 3]
NearIdentityInverse[x + x^Sqrt[2], {x, y}, 3]
```

Table is used so that the differentiation order is an actual nonnegative integer at each evaluation. This is a perturbation-depth formula. For general mixtures it does not promise an exponent-complete cutoff, and it provides no automatic branch or remainder checking. Its role is to expose the simple mathematical core of the construction.

B Derivation of the polynomial operator identity

For completeness, the general one-step rule is

$$\frac{d}{ds} \left[s^b Z \left(\frac{\log s}{p} \right) \right] = s^{b-1} \left(b + \frac{1}{p} D_L \right) Z(L).$$

Apply it a second time to obtain

$$s^{b-2} \left(b - 1 + \frac{1}{p} D_L \right) \left(b + \frac{1}{p} D_L \right) Z(L).$$

Induction yields (15). The crucial point is that the polynomial argument changes under differentiation; treating $\log s$ as a constant would omit every D_L term and produce wrong lower logarithmic coefficients. For example it would retain the Catalan top coefficient in the logarithmic example while giving $2L^2 + 4L + 2$, rather than $2L^2 + 5L + 3$, in the cubic block. A test checking only leading logarithms is therefore insufficient.

At fixed k , put $A_r = q + \Delta_k + pr$. Expanding the commuting product also gives

$$Q_k^{(q)}(L) = \frac{(-1)^{|k|} q}{p^{|k|} k!} \sum_{j=0}^{|k|-1} e_{|k|-1-j}(A_1, \dots, A_{|k|-1}) D_L^j \left(\prod_i P_i(L)^{k_i} \right).$$

This is an alternative finite coefficient formula. Both versions have the same polynomial degree bound; the sequential-operator form generally avoids constructing the elementary symmetric polynomials explicitly.

C Source provenance and scope of consultation

The source of the precise question was the user-supplied archive, whose Markdown text was read in full. The public question was also inspected to confirm the branch-selection issue and the irrational-power example [1]. Historical command outputs in that question are treated as historical observations, not as present-day benchmarks.

For the repository material, the directory listing, README, companion introduction, selected coefficient/remainder discussion, synthesis, and Wolfram recipe appendix were inspected through the GitHub connector. The large main *Transseries and Inversion* source was identified, but the available content/blob endpoints did not return its text successfully. Consequently no claim is made to have read that monograph in full. The present proofs are self-contained and do not depend on inaccessible portions.

The official Wolfram documentation was consulted for the roles of `InverseSeries`, `SeriesData`, `AsymptoticSolve`, and the test framework [5, 6, 7, 8, 9, 10]. Native behavior of the archived commands was not rerun. In particular, no blanket statement is made that current built-in asymptotic functions fail on all expressions in the class studied here.

References

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- [2] NIST Digital Library of Mathematical Functions, §1.10(iv), residue and Rouché theorems, and §1.10(vii), inverse functions and Lagrange inversion. Consulted September 7, 2026. <https://dlmf.nist.gov/1.10>.
- [3] Vladimir Reshetnikov / ProveIt repository, *Series and transseries*, directory README and file listing. Consulted September 7, 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries>.
- [4] ProveIt repository, *Combinatorial coefficient algebras for transseries inverses*, research draft, selected sections and appendices consulted September 7, 2026. https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries/Combinatorial_Transseries_Inverses/Combinatorial_Transseries_Inverses.tex.
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